

Experiment 10: Implement Expectation & Maximization Algorithm

Aim

To implement the Expectation-Maximization (EM) algorithm in Python for clustering data using a Gaussian Mixture Model (GMM).

Python Program

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.mixture import GaussianMixture

np.random.seed(42)

data1 = np.random.normal(5, 1, 100)
data2 = np.random.normal(10, 1, 100)
X = np.concatenate([data1, data2]).reshape(-1, 1)

model = GaussianMixture(
    n_components=2,
    random_state=42
)

model.fit(X)

labels = model.predict(X)

print("Means of Gaussian distributions:")
print(model.means_)

print("\nVariances:")
print(model.covariances_)

print("\nCluster probabilities:")
```


[illegible]